Bruise Segmentation Results Summary

Native-Argmax YOLO Evaluation and Subject-Level Analysis

Evaluation set: 185 white-light images from 28 subjects, evaluated against the 2-of-3 majority-vote mask on a common 640×640 grid.
YOLO evaluation: Native argmax only. Custom /255 YOLO results are excluded.

1. Overall Performance

Model	Parameters (M)	Mean Dice	Median Dice	Complete-miss (%)
SegFormer-B2 teacher	27.35	0.7692	0.8192	0.00
SegFormer-B0 distilled	3.71	0.7680	0.8167	0.00
SegFormer-B0 direct	3.71	0.7663	0.8129	0.54
YOLO26n distilled	1.63	0.7261	0.8012	2.16
YOLO26n direct	1.63	0.7021	0.8061	6.49
DeepLabV3+ (ResNet-50)	26.68*	0.7453 +/- 0.0182	0.8140	1.80 +/- 0.62

DeepLabV3+ is shown using the available aggregate result. Its 26.68 M parameter count corresponds to the standard ResNet-50 DeepLabV3+ implementation and should be checked against the exact saved model before publication.

Main observation

The three SegFormer models have nearly identical Dice scores. The clearest practical separation is the complete-miss rate: SegFormer-B2 and distilled B0 produce no complete misses in this evaluation, whereas native-argmax YOLO produces more blank masks. DeepLabV3+ lies within the same broad Dice range but has a 1.80% aggregate miss rate.

2. Paired Contrasts

Subject-level cluster bootstrap. Δ = model A − model B on the same resample. $P(\Delta > 0)$ is one-sided.

Contrast	Δ	95% CI	$P(\Delta > 0)$	Verdict
Distillation (B0-dist − B0-direct)	+0.0017	[-0.0091, +0.0126]	0.582	n.s.; no resolved distillation gain
Student − teacher (B0-dist − B2)	-0.0012	[-0.0196, +0.0176]	0.437	n.s.; no resolved difference
YOLO distill − direct	+0.0239	[-0.0142, +0.0590]	0.901	n.s.
SegFormer − YOLO (B0-dist − YOLO-direct)	+0.0659	[+0.0111, +0.1209]	0.994	significant
Miss rate: YOLO-direct − B0-dist	+6.49	[+2.39, +11.63]	1.000	significant

Interpretation

At 28 subjects, the Dice differences for distillation and for the compact student versus the teacher are smaller than the uncertainty that the study can resolve. The clearly resolved results are the SegFormer-versus-YOLO Dice contrast and the higher complete-miss rate of YOLO direct relative to SegFormer-B0 distilled.

3. Inference Performance

Timing boundary: 640×640 tensor already on the GPU through output-mask generation. Disk loading, image decoding, resizing, and host-to-device copy are excluded. YOLO uses native argmax.

Model	Median (ms)	p95 (ms)	FPS	Params (M)	Peak act. (MB)
SegFormer-B2 teacher	33.67	34.95	29.7	27.35	1779
SegFormer-B0 direct	16.68	16.94	59.9	3.71	1204
SegFormer-B0 distilled	16.64	17.07	60.1	3.71	1204
YOLO26n direct	8.18	8.36	122.2	1.63	967
YOLO26n distilled	8.22	8.51	121.7	1.63	967

- SegFormer-B0 provides approximately twice the throughput of SegFormer-B2 while using about 7.4 times fewer parameters.
- YOLO26n provides approximately twice the throughput of SegFormer-B0 and about four times the throughput of SegFormer-B2, but its complete-miss rate is higher.
- Direct and distilled versions have essentially identical deployment cost because distillation changes training rather than the deployed architecture.

4. Fairness Evaluation by ITA-Derived Skin-Tone Group

This analysis is exploratory. ITA is an image-derived skin-tone proxy and should not be interpreted as race, ethnicity, or self-identified skin tone.

Model	KW p	Sig.	Dice gap	Best group	Worst group	Miss gap (pp)
SegFormer-B2 teacher	0.028	Yes	0.094	Intermediate (III-IV)	Tan (IV)	0.00
SegFormer-B0 direct	0.709	No	0.064	Intermediate (III-IV)	Dark (VI)	2.56
SegFormer-B0 distilled	0.749	No	0.043	Light (II-III)	Dark (VI)	0.00
YOLO26n direct	0.212	No	0.131	Brown (V)	Light (II-III)	15.38
YOLO26n distilled	0.063	No	0.075	Brown (V)	Light (II-III)	7.69

Only SegFormer-B2 shows a significant omnibus difference ($p = 0.028$). After Bonferroni correction, the Intermediate-versus-Dark comparison remains significant (adjusted $p = 0.0388$). The other four models do not show a significant omnibus difference. Native-argmax YOLO has its lowest median Dice and largest miss rate in the Light group, but the subgroup analysis is underpowered and should not be presented as a confirmed fairness failure.

Per-group results

Model	ITA group	Images	Median Dice	Mean recall	Miss (%)
SegFormer-B2 teacher	Light (II-III)	39	0.789	0.734	0.00
SegFormer-B2 teacher	Intermediate (III-IV)	38	0.867	0.786	0.00
SegFormer-B2 teacher	Tan (IV)	24	0.773	0.677	0.00
SegFormer-B2 teacher	Brown (V)	29	0.810	0.729	0.00
SegFormer-B2 teacher	Dark (VI)	55	0.793	0.706	0.00
SegFormer-B0 direct	Light (II-III)	39	0.827	0.741	2.56
SegFormer-B0 direct	Intermediate (III-IV)	38	0.834	0.762	0.00
SegFormer-B0 direct	Tan (IV)	24	0.810	0.684	0.00
SegFormer-B0 direct	Brown (V)	29	0.813	0.742	0.00
SegFormer-B0 direct	Dark (VI)	55	0.770	0.704	0.00
SegFormer-B0 distilled	Light (II-III)	39	0.829	0.710	0.00
SegFormer-B0 distilled	Intermediate (III-IV)	38	0.819	0.757	0.00
SegFormer-B0 distilled	Tan (IV)	24	0.795	0.681	0.00
SegFormer-B0 distilled	Brown (V)	29	0.822	0.747	0.00
SegFormer-B0 distilled	Dark (VI)	55	0.786	0.715	0.00
YOLO26n direct	Light (II-III)	39	0.714	0.516	15.38
YOLO26n direct	Intermediate (III-IV)	38	0.828	0.698	5.26
YOLO26n direct	Tan (IV)	24	0.778	0.667	0.00
YOLO26n direct	Brown (V)	29	0.845	0.735	6.90
YOLO26n direct	Dark (VI)	55	0.787	0.726	3.64
YOLO26n distilled	Light (II-III)	39	0.758	0.581	7.69
YOLO26n distilled	Intermediate (III-IV)	38	0.827	0.763	0.00
YOLO26n distilled	Tan (IV)	24	0.813	0.698	0.00
YOLO26n distilled	Brown (V)	29	0.833	0.752	0.00
YOLO26n distilled	Dark (VI)	55	0.772	0.639	1.82

5. Final Takeaway

SegFormer-B0 distilled provides nearly the same Dice as the B2 teacher at approximately twice the inference throughput and with far fewer parameters. The paired analysis does not establish a Dice improvement from distillation, so its main value here is compact deployment with a very low complete-miss rate. Native-argmax YOLO is substantially faster, but its higher complete-miss rate remains the principal limitation. DeepLabV3+ is a competitive external baseline and should be presented descriptively because paired per-image contrasts against the five core models are not available in the supplied result package.